

HDOC – Day 22

C Programming Assignment

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Problem: Accept two positive integers m and n where $m \leq n$. Repeatedly display a menu and use a switch statement to perform multiplication tables, a multiplication grid, consecutive-integer output, or exit. Invalid menu choices must produce an error message.

Algorithm

1. Start.
2. Read positive integers m and n .
3. If $m \leq 0$, $n \leq 0$, or $m > n$, display an invalid-input message and stop.
4. Use a do-while loop to display the menu until Exit is selected.
5. Read the menu choice and use a switch statement.
6. Choice 1: for $i = 1$ to 10, print $m \times i$ and $n \times i$ side by side.
7. Choice 2: print column headings 1 to n ; for each row $i = 1$ to m , print $i \times j$ for $j = 1$ to n .
8. Choice 3: for each i from m to n , print i followed by $i+1$, $i+2$, $i+3$, $i+4$, and $i+5$.
9. Choice 4: display the exit message and terminate the loop.
10. Default: display an invalid menu choice message, then show the menu again.
11. Stop.

Test-Case Table

TC	m	n	Choice	Expected Result	Status
1	3	5	1	Tables of 3 and 5 from multiplier 1 to 10	Pass
2	3	5	2	3 x 5 multiplication grid	Pass
3	3	5	3	Rows for 3, 4, 5; each followed by next five integers	Pass
4	3	5	7	Invalid menu choice message; menu shown again	Pass
5	0	5	-	Invalid input message because m is not positive	Pass

C Program

```
#include <stdio.h>
int main(void) { int
m, n, choice, i, j;
printf("Enter two positive integers m and n (m <= n): ");
scanf("%d %d", &m, &n);
if (m <= 0 || n <= 0 || m > n) {
printf("Invalid input. Please enter positive integers with m <= n.\n"); return 0;
}
do {
printf("\n1. Multiplication tables of m and n\n");
printf("2. Multiplication grid\n"); printf("3.
Consecutive integers\n");
printf("4. Exit\n"); printf("Enter your choice: "); scanf("%d", &choice); switch
(choice) { case 1: printf("\nMultiplication Tables of %d and %d (m=%d,
n=%d)\n", m, n, m, n); for (i = 1; i <= 10; i++)
printf("%d x %d = %-4d %d x %d = %d\n", m, i, m*i, n, i, n*i); break;
case 2: printf("\nMultiplication Grid (m=%d, n=%d)\n",
m, n); printf(" ");
for (j = 1; j <= n; j++) printf("%4d", j); printf("\n
");
```

```

for (j = 1; j <= n; j++) printf("----");
printf("\n"); for (i = 1; i <= m; i++)
{ printf("%2d |", i);
for (j = 1; j <= n; j++) printf("%4d", i*j); printf("\n");
}
break; case
3:
printf("\nConsecutive Integers from %d to %d (m=%d, n=%d)\n", m, n, m, n);
for (i = m; i <= n; i++) { printf("%d:", i);
for (j = 0; j <= 5; j++) printf(" %d", i+j); printf("\n");
}
break; case 4: printf("Exiting program.\n"); break; default:
printf("Invalid menu choice. Please select 1, 2, 3, or 4.\n");
}
} while (choice != 4);
return 0;
}

```

Compilation, Execution and Testing

```
Session Actions Edit View Help
(kaliⓀkali)~[~]
$ vi pro.c

(kaliⓀkali)~[~]
$ cc pro.c

(kaliⓀkali)~[~]
$ ./a.out
Enter two positive integers m and n (m ≤ n): 3 5
```

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 1

Multiplication Tables of 3 and 5 (m=3, n=5)

3 x 1 = 3	5 x 1 = 5
3 x 2 = 6	5 x 2 = 10
3 x 3 = 9	5 x 3 = 15
3 x 4 = 12	5 x 4 = 20
3 x 5 = 15	5 x 5 = 25
3 x 6 = 18	5 x 6 = 30
3 x 7 = 21	5 x 7 = 35
3 x 8 = 24	5 x 8 = 40
3 x 9 = 27	5 x 9 = 45
3 x 10 = 30	5 x 10 = 50

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 2

Multiplication Grid (m=3, n=5)

	1	2	3	4	5
1	1	2	3	4	5
2	2	4	6	8	10
3	3	6	9	12	15

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 3

Consecutive Integers from 3 to 5 (m=3, n=5)

3: 3 4 5 6 7 8
4: 4 5 6 7 8 9
5: 5 6 7 8 9 10

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 7

Invalid menu choice. Please select 1, 2, 3, or 4.

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 4

Exiting program.

```
(kali@kali)-[~]
```

```
$ ./a.out
```

Enter two positive integers m and n ($m \leq n$): 0 5

Invalid input. Please enter positive integers with $m \leq n$.

```
(kali@kali)-[~]
```